

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

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FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 1.2: MECHANICAL PROPERTIES OF MATERIALS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION — STUDENT ASSESSMENT DOCUMENT

Unit: Mechanical Properties of Materials | Grade 10 Physics

Sub-strand 1.2 | Lessons 1–6

WHAT IS THIS DOCUMENT?

This is your Final Explanation — the culminating task for the entire unit. You will use everything you have learned across all 6 lessons to answer the driving question:

"Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?"

Think back to the Kericho farmer's suspension footbridge. Two sisal ropes looked almost identical, yet one bounced back and the other snapped under the weight of an

80 kg sack of tea leaves. By the end of this explanation, you should be able to say — with evidence, equations, and scientific reasoning — exactly why that happened and how we could have predicted it.

HOW TO USE THIS DOCUMENT

- Work individually. You have 80–100 minutes.
- Use your Summary Table and your Final Model diagram as reference tools.
- Answer every section in order — each section builds on the last.
- Write in full sentences and paragraphs unless a calculation box is shown.
- Show ALL working for every calculation. Correct units are required for full marks.
- Connect your answers back to the Kericho bridge phenomenon wherever you can — this shows deep understanding.
- You may draw labelled diagrams to support your written explanations.

GRADING BREAKDOWN

- Scientific Understanding 25%
- Use of Evidence 25%
- Reasoning & Calculations 20%
- Communication & Vocabulary 15%
- Real-World Application 15%

WRITING TIPS

- ✓ Use correct scientific vocabulary in every paragraph (elasticity, plastic deformation, Hooke's Law, elastic limit, stress, strain, Young's modulus, ultimate tensile strength).
- ✓ Name specific materials you tested — do not say "the material"; say "the rubber band," "the copper wire," or "the sisal rope."
- ✓ Explain WHY and HOW, not just WHAT. "The rope snapped because..." is better than "The rope snapped."
- ✓ Show that your thinking GREW from Lesson 1 to Lesson 6. You can use phrases like "At first I thought... but after Lesson 3 I understood that..."
- ✓ If you are unsure how to start a paragraph, try this stem: "The evidence from [Lesson X] shows that... This connects to the bridge because..."

PART 1 — Mechanical Properties: Understanding How Materials Behave

In this section, explain the key mechanical properties of materials that you investigated during Lesson 1. Your explanation must cover ALL of the following properties:

During Lesson 1, I investigated how different materials respond when a force is applied to them. I learned that materials can be classified by their mechanical properties — the characteristics that describe how they behave under stress.

Elasticity is the ability of a material to return to its original shape and size after a deforming force is removed. At the Lesson 1 exploration station, I stretched a rubber band and then released it — it snapped back to its original length. This is elastic behaviour.

elasticity, plasticity, and brittleness; ductility and malleability; strength, hardness, and toughness. For EACH property, you must: (a) define it clearly in your own words, (b) give at least one Kenyan or locally relevant example, and (c) explain why it matters in real life — especially for structures or tools used in Kenya. Then connect your explanation back to the Kericho bridge: Which mechanical property best explains why one rope bounced back and the other did not? Which property explains why the second rope snapped rather than simply deforming permanently? Finally, describe the specific evidence from the Lesson 1 exploration stations. What did you observe when you tested rubber bands, clay, a dry spaghetti strand (or glass rod), and copper wire? What did each test tell you about the material's properties?	The new sisal rope on the Kericho farmer's bridge showed elasticity: it stretched under the 80 kg load of tea leaves and returned to its original length when the load was removed. Plasticity is the opposite — a material that is permanently deformed after a force is applied. When I pressed clay (similar to the red soil found near Kisumu), it stayed in its new shape. The old, weathered sisal rope on the bridge showed plastic behaviour before it snapped: it did not return to its original length even before breaking. Brittleness means a material breaks suddenly with little or no plastic deformation. When I snapped a dry spaghetti strand, it did not bend or stretch — it fractured cleanly. Glass, used in windows of Nairobi office buildings, is another example: it shatters without warning rather than bending. Ductility is the ability of a material to be drawn into a thin wire without breaking. Copper is highly ductile — the electrical wiring in Kenyan homes is copper drawn into long, thin wires. Malleability is the ability to be hammered or rolled into sheets. Aluminium is malleable — it is pressed into the thin sheets used to make sufuria (cooking pots) sold in markets across Kenya. Although both ductility and malleability involve permanent shape change, ductility involves pulling and malleability involves compressing. Strength is a material's ability to withstand large forces without breaking. Hardness is resistance to scratching or surface indentation. Toughness is the ability to absorb energy and resist fracturing — a tough material may dent but will not shatter. These three properties are related but distinct: steel used in the columns of buildings along Moi Avenue, Nairobi, must be strong AND tough — it must carry huge loads AND resist sudden impacts without cracking. The Kericho bridge puzzle is now clearer. The new rope was elastic: it stored energy and released it, returning to its original length. The old rope had lost its elasticity — possibly due to weathering, repeated overloading beyond its elastic limit, or fibre degradation. Once its elastic limit was exceeded, it deformed plastically and then reached its breaking stress, causing it to snap. This behaviour relates to strength: the old rope's tensile strength had been reduced by wear. If we had known the breaking stress of both ropes beforehand, we could have predicted the failure.
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PART 2 — Hooke's Law: The Relationship Between Force and Extension

In this section, explain what Hooke's Law states and what it means scientifically. Your explanation must include: (a) the equation $F = kx$ with every variable defined and its SI unit stated; (b) what the spring constant k tells us about a material or spring; (c) the elastic limit — what it is, what	Hooke's Law describes the behaviour of elastic materials in their linear region. It states that the extension of a spring (or elastic material) is directly proportional to the applied force, provided the elastic limit is not exceeded. The equation is: $F = kx$ where F is the applied force in Newtons (N), x is the extension in metres (m), and k is the spring constant in Newtons per metre (N/m). The spring constant k tells us how stiff a spring or material is. A large value of k means the material is very stiff — it requires a large force to produce even a small extension. A small value of k means the material is more flexible.
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happens to the material if force exceeds it, and how it appears on a force-extension graph; (d) a description of the shape of a force-extension graph, including the linear (Hookean) region and the post-elastic-limit region; and (e) how elastic materials store energy (you may refer to the area under the force-extension graph).

Then include your experimental evidence from Lesson 2. Describe what you measured, show your data, and explain what it proves.

Finally, complete the two calculation boxes below using the values given. Show all working and state the meaning of your answer.

The elastic limit is the maximum force (or stress) up to which a material obeys Hooke's Law and will return to its original shape. If the force exceeds the elastic limit, the material undergoes plastic deformation — it will NOT return to its original shape when the load is removed. For the old sisal rope in Kericho, years of repeated heavy loads likely pushed the rope fibres past their elastic limit many times. Each time, the rope stayed slightly more deformed than before. By the day the 80 kg sack was placed on it, the rope had very little elastic capacity left.

On a force-extension graph, the Hookean (elastic) region is a straight line passing through the origin — this is where $F = kx$ holds. The gradient of this straight line equals the spring constant k . Beyond the elastic limit, the graph curves away from the straight line: a much larger extension is produced for each additional unit of force, and the material will not recover. Eventually, the graph reaches the breaking point, where the material fractures.

Elastic materials store energy when stretched. This stored energy is called elastic potential energy. On a force-extension graph, the elastic potential energy stored equals the area under the straight-line (Hookean) region: $E = \frac{1}{2}Fx = \frac{1}{2}kx^2$. This is why a stretched rubber band can flick an object when released — it converts stored elastic potential energy into kinetic energy.

In Lesson 2, I hung calibrated masses on a steel spring and measured the extension with a ruler. My results showed:

— 1 N load → 2.5 cm extension (0.025 m)

— 2 N load → 5.0 cm extension (0.050 m)

— 3 N load → 7.5 cm extension (0.075 m)

Doubling the force exactly doubled the extension each time, confirming Hooke's Law in the elastic region. When I plotted a force-extension graph, I got a straight line through the origin.

CALCULATION BOX A — Spring Constant

Given: A spring used in a school physics lab in Nairobi extends by 6.0 cm (0.060 m) when a force of 3.0 N is applied. Find the spring constant k .

Working:

$$F = kx$$

$$k = F \div x$$

$$k = 3.0 \text{ N} \div 0.060 \text{ m}$$

$$k = 50 \text{ N/m}$$

Interpretation: This spring requires 50 N of force to extend it by 1 metre. It is moderately stiff.

CALCULATION BOX B — Predicting Extension

Given: A second spring has a spring constant of $k = 80 \text{ N/m}$. A student applies a force of 4.0 N. What is the extension, assuming

	the elastic limit is not exceeded? Working: $F = kx \rightarrow x = F \div k$ $x = 4.0 \text{ N} \div 80 \text{ N/m}$ $x = 0.050 \text{ m} = 5.0 \text{ cm}$ Interpretation: The stiffer spring ($k = 80 \text{ N/m}$) extends only 5.0 cm under the same force that might extend a softer spring much further. For bridge ropes, a higher k means less sag under load — an important design factor.
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PART 3 — Stress and Strain: Measuring Material Response Independently of Size

In this section, explain why scientists use stress and strain instead of just force and extension when comparing materials. Your explanation must include: (a) the definition and equation for tensile stress ($\sigma = F/A$), with units; (b) the definition and equation for tensile strain ($\epsilon = \Delta L/L_0$), explaining why it has no units; (c) the concept of breaking stress (ultimate tensile stress) and why it is a material property rather than an object property; and (d) a brief explanation of how stress and strain allow us to compare a thin sisal rope and a thick steel cable fairly. Then describe your experimental evidence from Lesson 3. Finally, complete the three calculation boxes below. Show all working and explain what each answer tells you about the material.	In Lesson 2, I noticed that a thin rubber band stretched much more than a thick one under the same force. This is a problem: force and extension alone depend on the SIZE and SHAPE of the object, not just the material itself. To compare materials fairly — regardless of their dimensions — scientists use stress and strain. Tensile stress (σ) measures how much force is applied per unit of cross-sectional area. The equation is: $\sigma = F / A$ where F is the applied force in Newtons (N) and A is the cross-sectional area in square metres (m^2). The unit of stress is the Pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$. Stress tells us how 'concentrated' the force is inside the material. A thin rope carrying the same load as a thick rope experiences much higher stress — this is why thin ropes snap first. Tensile strain (ϵ) measures how much a material stretches relative to its original length. The equation is: $\epsilon = \Delta L / L_0$ where ΔL is the change in length (extension) and L_0 is the original length, both in the same unit. Because it is a ratio of two lengths, strain has NO unit — it is a pure number (sometimes expressed as a percentage). Strain tells us the fractional deformation of the material. Breaking stress (ultimate tensile stress, UTS) is the maximum stress a material can withstand before it fractures. It is a property of the MATERIAL — not of any particular rope or wire. This is crucial: if we know the UTS of old sisal fibre, we can calculate in advance whether any rope made of that fibre (whatever its thickness) will snap under a given load. For the Kericho farmer, testing the breaking stress of the old rope's sisal fibre before building the bridge could have prevented the accident. In Lesson 3, I tested a copper wire with original length $L_0 = 2.0 \text{ m}$ and cross-sectional area $A = 1.0 \times 10^{-6} \text{ m}^2$. I hung a 10 N weight and measured an extension $\Delta L = 0.002 \text{ m}$.
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CALCULATION BOX C — Stress and Strain (Copper Wire)

Given: $F = 10 \text{ N}$, $A = 1.0 \times 10^{-6} \text{ m}^2$, $L_0 = 2.0 \text{ m}$, $\Delta L = 0.002 \text{ m}$

Stress: $\sigma = F/A = 10 \div (1.0 \times 10^{-6}) = 1.0 \times 10^7 \text{ Pa} = 10 \text{ MPa}$

Strain: $\epsilon = \Delta L/L_0 = 0.002 \div 2.0 = 0.001$ (no units)

Interpretation: The copper wire experiences 10 MPa of stress and deforms by 0.1% of its original length. This is within the elastic region for copper.

CALCULATION BOX D — Stress on a Sisal Rope

Given: The new sisal rope on the Kericho bridge has a cross-sectional area of $4.0 \times 10^{-4} \text{ m}^2$. The 80 kg sack of tea leaves applies a force equal to its weight (use $g = 10 \text{ m/s}^2$).

Step 1 — Find force: $F = mg = 80 \times 10 = 800 \text{ N}$

Step 2 — Find stress: $\sigma = F/A = 800 \div (4.0 \times 10^{-4}) = 2.0 \times 10^6 \text{ Pa} = 2.0 \text{ MPa}$

Interpretation: The rope experiences 2.0 MPa of tensile stress. If the breaking stress of new sisal is approximately 150 MPa, the rope is well within its safe range — consistent with the observation that the new rope did NOT snap.

CALCULATION BOX E — Stress on a Steel Wire

Given: A steel support wire in a Nairobi construction project has cross-sectional area $A = 2.0 \times 10^{-6} \text{ m}^2$, original length $L_0 = 3.0 \text{ m}$, applied force $F = 50 \text{ N}$, and measured extension $\Delta L = 0.003 \text{ m}$.

Stress: $\sigma = F/A = 50 \div (2.0 \times 10^{-6}) = 2.5 \times 10^7 \text{ Pa} = 25 \text{ MPa}$

Strain: $\epsilon = \Delta L/L_0 = 0.003 \div 3.0 = 0.001$ (no units)

Interpretation: The steel wire is under 25 MPa stress with only 0.1% strain — very small deformation, consistent with steel's high stiffness.

PART 4 — Young's Modulus: The Stiffness Fingerprint of a Material

In this section, explain what Young's modulus is and why it is one of the most useful single

Young's modulus (E) is a measure of the stiffness of a material — how much it resists deformation when stress is applied. It is defined as the ratio of tensile stress to tensile strain within the elastic (Hookean) region:

numbers in materials engineering. Your explanation must include: (a) the definition and equation $E = \sigma / \epsilon = (F/A) \div (\Delta L/L_0)$, with all variables defined and units stated; (b) what a high versus low Young's modulus tells us about a material's stiffness; (c) how Young's modulus is found from a stress-strain graph (explain the gradient); (d) comparison of at least three materials by their Young's modulus values, using Kenyan engineering or everyday contexts; and (e) how Young's modulus relates back to the Kericho bridge — specifically, how comparing the Young's modulus of new and old sisal fibre could help predict rope failure.

Then show the full calculation for Young's modulus using the steel wire values from Calculation Box E (Part 3).

Finally, describe how your understanding of Young's modulus grew from Lesson 4 and how you used the simulation or stress-strain graph activity to confirm your understanding.

$$E = \sigma / \epsilon \quad \text{or equivalently} \quad E = (F \times L_0) / (A \times \Delta L)$$

The unit of Young's modulus is the Pascal (Pa), because strain is dimensionless. In practice, Young's modulus values are very large, so they are often quoted in gigapascals (GPa), where $1 \text{ GPa} = 1 \times 10^9 \text{ Pa}$.

A high Young's modulus means a material is very stiff — a large stress produces only a tiny strain. Steel has $E \approx 200 \text{ GPa}$; this is why steel beams used in the construction of Nairobi's expressway barely bend under enormous loads. A low Young's modulus means a material is flexible — a small stress produces a large strain. Rubber has $E \approx 0.01\text{--}0.1 \text{ GPa}$; this is why rubber shock absorbers in matatu suspensions deform easily under bumps but do not break.

On a stress-strain graph, the gradient (slope) of the straight-line (elastic) region equals Young's modulus:

$$E = \text{gradient} = \Delta \sigma / \Delta \epsilon$$

A steeper gradient means a higher E and a stiffer material. During Lesson 4, I plotted stress-strain graphs for copper and steel wires and confirmed that the steel graph had a steeper gradient, consistent with steel's higher Young's modulus.

Comparison of materials in Kenyan contexts:

- Steel ($E \approx 200 \text{ GPa}$): Used in reinforcing bars (rebar) in concrete buildings and bridges across Kenya. Very stiff, high strength.
- Copper ($E \approx 120 \text{ GPa}$): Used in electrical wiring throughout Kenya Power infrastructure. Stiffer than aluminium but more ductile.
- Sisal fibre ($E \approx 15\text{--}20 \text{ GPa}$): A natural fibre grown commercially in the Rift Valley and Taita Hills regions of Kenya. Flexible enough for rope-making, but much lower stiffness than metals.
- Rubber ($E \approx 0.01 \text{ GPa}$): Used in tyres, seals, and shock absorbers. Highly elastic with very low stiffness.

For the Kericho bridge: new sisal rope and old weathered sisal rope are both made of sisal fibre, yet they behaved differently. Weathering, UV exposure, and repeated overloading degrade the long molecular chains in sisal, reducing both its Young's modulus AND its breaking stress. If we had tested both ropes by measuring stress and strain, we would have found that the old rope had a lower E (more deformation per unit stress) and a lower breaking stress — giving us an early warning before it snapped.

CALCULATION BOX F — Young's Modulus (Steel Wire)

Using values from Calculation Box E:

$$\sigma = 25 \text{ MPa} = 2.5 \times 10^7 \text{ Pa}$$

$$\epsilon = 0.001$$

$$E = \sigma / \epsilon = (2.5 \times 10^7) \div 0.001 = 2.5 \times 10^{10} \text{ Pa} = 25 \text{ GPa}$$

Note: Real steel has $E \approx 200 \text{ GPa}$. The lower value here reflects the simplified experimental data used in this exercise. The calculation method is correct.

Interpretation: Young's modulus gives us a single number that characterises the stiffness of the steel — independent of how thick or long the wire is. Any steel wire, regardless of its dimensions, will have the same E value. This is why engineers in Kenya can look up E in a materials databook and use it to design safe structures without testing every individual piece of steel.

PART 5 — Bringing It All Together: Answering the Driving Question and Applying Your Knowledge

This is your most important section. You must now use ALL of the concepts from Parts 1–4 to write a complete, integrated answer to the driving question:

'Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?'

Your answer must do ALL of the following:

(a) EXPLAIN THE KERICHO BRIDGE PHENOMENON FULLY — using elasticity, elastic limit, Hooke's Law, stress, strain, and breaking stress to explain exactly what happened to each rope and why.

(b) DESCRIBE HOW YOUR THINKING CHANGED — what did you think at the start of Lesson 1, and what do you understand now that you did not understand before? Be specific.

(c) APPLY YOUR KNOWLEDGE TO A NEW KENYAN CONTEXT — choose ONE of the following and explain, using the concepts

(a) FULL EXPLANATION OF THE KERICHO BRIDGE

The two sisal ropes on the Kericho farmer's footbridge looked similar but were fundamentally different in their mechanical properties. The new rope had its molecular structure intact — the long sisal fibres were undamaged, well-aligned, and capable of storing elastic potential energy when stretched. When the 80 kg sack of tea leaves (weight = 800 N) was placed on the bridge, the new rope experienced a tensile stress of approximately 2.0 MPa (as calculated in Part 3). This was well below its elastic limit. In the elastic region, the rope obeyed Hooke's Law: extension was proportional to force. When the load was removed, the fibres released their stored elastic potential energy and the rope returned to its original length — elastic behaviour.

The old rope told a completely different story. Years of heavy loads, weather exposure (rain, UV, humidity from the Kericho highlands), and possible insect damage had broken down the sisal fibres at the molecular level. Its elastic limit had dropped significantly — meaning even moderate stress pushed it into plastic deformation. Its spring constant k was lower (it extended more per newton of applied force), and its breaking stress (ultimate tensile stress) had been greatly reduced. When the same 800 N load was applied, the stress in the old rope exceeded its already-reduced breaking stress. The fibres failed progressively — first stretching plastically (not returning to original length), then fracturing completely. The rope snapped.

In summary: the difference was not about appearance — it was about mechanical properties. The new rope had high elasticity, a high elastic limit, and a high breaking stress. The old rope had low elasticity, a low elastic limit, and a low breaking stress.

(b) HOW MY THINKING CHANGED

At the start of Lesson 1, I thought that whether a rope snapped depended mostly on how thick it was — a thicker rope would always be stronger. I also thought that 'stretching' and 'breaking' were just two stages of the same thing. I did not understand that materials have distinct properties (elasticity, plasticity, brittleness) that are independent of size, or that scientists can measure and predict these properties mathematically.

By Lesson 3, I understood that stress and strain allow us to compare materials regardless of their dimensions — a key insight that changed how I thought about the bridge. By Lesson 4, I could calculate Young's modulus and understood that it is a fingerprint of the material, not the object. By Lesson 6, I can now explain the entire bridge failure using equations, graphs, and scientific vocabulary — and I can design a test to predict whether a rope will fail before it is ever used on a bridge.

(c) APPLICATION — Option C: Designing Suspension Ropes for a Tea-Transport Aerial Ropeway in Kericho

A materials engineer designing suspension ropes for an aerial ropeway on a Kericho tea farm would follow this process:

from this unit, how a materials engineer would use mechanical properties data to make a safe design decision: • Option A: Selecting cables for a new footbridge over the Tana River in Garissa County. • Option B: Choosing reinforcing steel (rebar) for a school building being constructed in Kisumu. • Option C: Designing the suspension ropes for a new tea-transport aerial ropeway on a farm in Kericho. (d) REFER TO YOUR FINAL MODEL — describe what your final model shows and how it is different from your Lesson 1 model. (e) REFLECT ON THE INQUIRY QUESTIONS — answer both KICD key inquiry questions directly: 1. What determines how a material responds when a force is applied to it? 2. How are the mechanical properties of materials applied in engineering and everyday life in Kenya?	Step 1 — Calculate the maximum load. If the ropeway must carry 120 kg of tea in each basket, the maximum force per rope is $F = mg = 120 \times 10 = 1,200 \text{ N}$. A safety factor of at least $5\times$ is standard in Kenya's structural engineering guidelines, so the rope must be designed to withstand 6,000 N without breaking. Step 2 — Determine required breaking stress. If the engineer wants to use steel cables with cross-sectional area $A = 1.0 \times 10^{-5} \text{ m}^2$, the stress under maximum design load is $\sigma = F/A = 6,000 \div (1.0 \times 10^{-5}) = 6.0 \times 10^8 \text{ Pa} = 600 \text{ MPa}$. The engineer would consult a materials databook and confirm that the chosen steel grade has a UTS (ultimate tensile stress) well above 600 MPa — which standard structural steel (UTS $\approx 400\text{--}600 \text{ MPa}$) may not satisfy, suggesting the engineer should specify high-tensile steel (UTS $\approx 800\text{--}1,000 \text{ MPa}$). Step 3 — Check for acceptable elastic deformation. Using Young's modulus for steel ($E \approx 200 \text{ GPa}$) and the working stress ($\sigma = 120 \text{ MPa}$ at normal load), the engineer calculates strain $\epsilon = \sigma/E = 120 \times 10^6 \div 200 \times 10^9 = 6.0 \times 10^{-4}$. For a rope of length $L_0 = 50 \text{ m}$, the extension $\Delta L = \epsilon \times L_0 = 6.0 \times 10^{-4} \times 50 = 0.03 \text{ m} = 3 \text{ cm}$. This small elastic sag is acceptable. Step 4 — Specify inspection intervals. Because the Kericho highlands have high humidity and rainfall, the engineer would also specify galvanised (rust-protected) steel and regular inspection for plastic deformation or kinking — early signs that the elastic limit has been exceeded. This systematic use of stress, strain, Young's modulus, and breaking stress is exactly what separates a scientifically designed structure from a guessed one — and it is what could have saved the Kericho farmer's bridge. (d) MY FINAL MODEL My Lesson 1 model showed a rope simply getting longer when a weight was hung on it, with an arrow pointing down for force and an arrow pointing up labelled 'rope holds it.' It had no labels for stress, strain, elastic limit, or molecular structure. My Final Model now shows: (1) a force-extension graph with the elastic region (straight line, gradient = k), the elastic limit point, and the plastic/fracture region labelled; (2) a stress-strain graph with gradient = Young's modulus E; (3) a molecular diagram showing aligned fibres in new sisal (capable of elastic recovery) versus damaged, unaligned fibres in old sisal (unable to recover); (4) calculations linking the 800 N load to the stress on each rope, and comparison to their respective breaking stresses. The model tells the complete story that my Lesson 1 drawing could not. (e) ANSWERING THE KICD KEY INQUIRY QUESTIONS Question 1 — What determines how a material responds when a force is applied to it? A material's response to force is determined by its mechanical properties, which in turn arise from its internal molecular and structural make-up. Specifically: its spring constant (k) determines how much it extends per unit force in the elastic region; its elastic limit determines the maximum stress it can experience and still recover; its Young's modulus (E) determines its stiffness (stress-to-strain ratio in the elastic region); and its ultimate tensile stress (breaking stress) determines the maximum force per unit area it can withstand before fracturing. Materials with intact, well-organised internal structures (like new sisal or undamaged steel) tend to have higher values of all these quantities. Degraded, weathered, or overloaded materials have lower values — making failure more likely.
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Question 2 — How are the mechanical properties of materials applied in engineering and everyday life in Kenya?

Mechanical properties data is used every day by engineers across Kenya. Civil engineers building schools, hospitals, and bridges in counties from Turkana to Mombasa select steel grades based on published Young's modulus and UTS values to ensure structures will not deform or collapse under expected loads. Farmers and rural communities who build footbridges, grain stores, and water-harvesting structures benefit from understanding that older, weathered ropes and beams have reduced mechanical properties — and should be replaced before they fail. Manufacturers of matatu body panels choose materials based on their ductility and toughness, while road builders in the Rift Valley consider the hardness and strength of stone aggregate. In every case, the scientific framework of stress, strain, elasticity, and Young's modulus allows quantitative, evidence-based decisions — replacing guesswork with engineering.